



000000000000 · 00000 18 00 · INT6068 Neural Networks and Deep Learning



▪ Agenda

01



Data Analysis

15%

~3 min

02



Data Mining

15%

~3 min

03



Course Design

35%

~6 min

04



Activities

35%

~6 min

PART 01



Data Analysis · 15% · ~3 min

01

🚲🚲🚲 — UCI Bike Sharing Dataset

17,379 🚲

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2011 - 2012

📅📅📅 · 📅📅📅 Capital Bikeshare

13 📅📅

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70 : 30

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📅📅📅📅

📅📅📅📅📅 2011-2012 📅

📅📅 Dashboard 📅📅📅

□□□□□□□□ ▪ Key Insights

□□□□□□

□□□□□□□□ $r \approx 0.40$ □□□□□□□□ 20-35°C □
□□ 38°C □□□□□□



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□□□□□□□□ 8 □□□□□□□□ 17-18 □□□□□□□□ 10-
15 □□□□□□□□

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2012 □□□□□□□□ 2011 □□□□□□ 64% □□□□□□□□
80% □□□□□□□□

PART 02



Data Mining · 15% · ~3 min

02

Decision Tree

Root Node

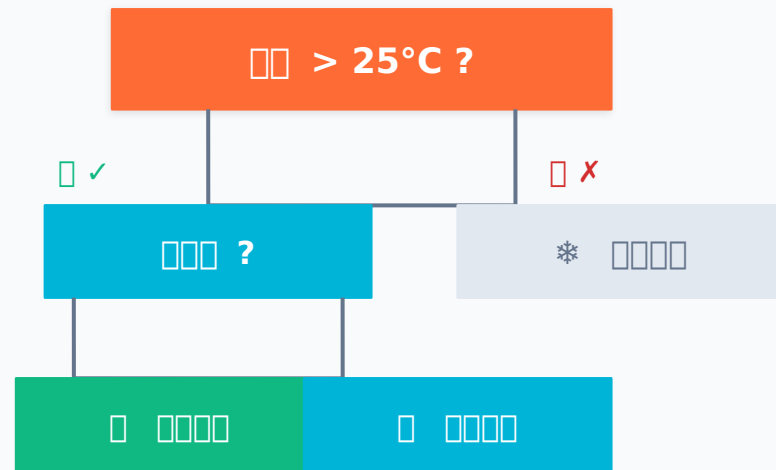
Is temperature
greater than 25°C?

Branches

Yes / No
Decision

Leaf Node

Final
Decision



Final Decision = Final Decision

□□□□□□□□□□

□□□□□

□□□□ R^2 & RMSE □□□□

□□□□ Summary □□□□□□

□□□□

□ **XGBoost**
 $R^2 \approx 0.950$ · RMSE ≈ 53.4

□ □□□□
 $R^2 \approx 0.930$ · RMSE ≈ 63.2

□ □□□□
 $R^2 \approx 0.920$ · RMSE ≈ 67.8

4 □□□
 $R^2 \approx 0.730$ · RMSE ≈ 124

5 □□□□
 $R^2 \approx 0.730$ · RMSE ≈ 124

No Free Lunch □□□□□□□□□□□□□□□□
□□□□□□□□□□

PART 03



Course Design · 35% · ~6 min

03

STEM 學習 · Learning Outcomes

科學 · 物理

- 理解科學探究的過程
- 理解科學 vs 技術

數學

- 理解數學的應用
- 理解數學的邏輯

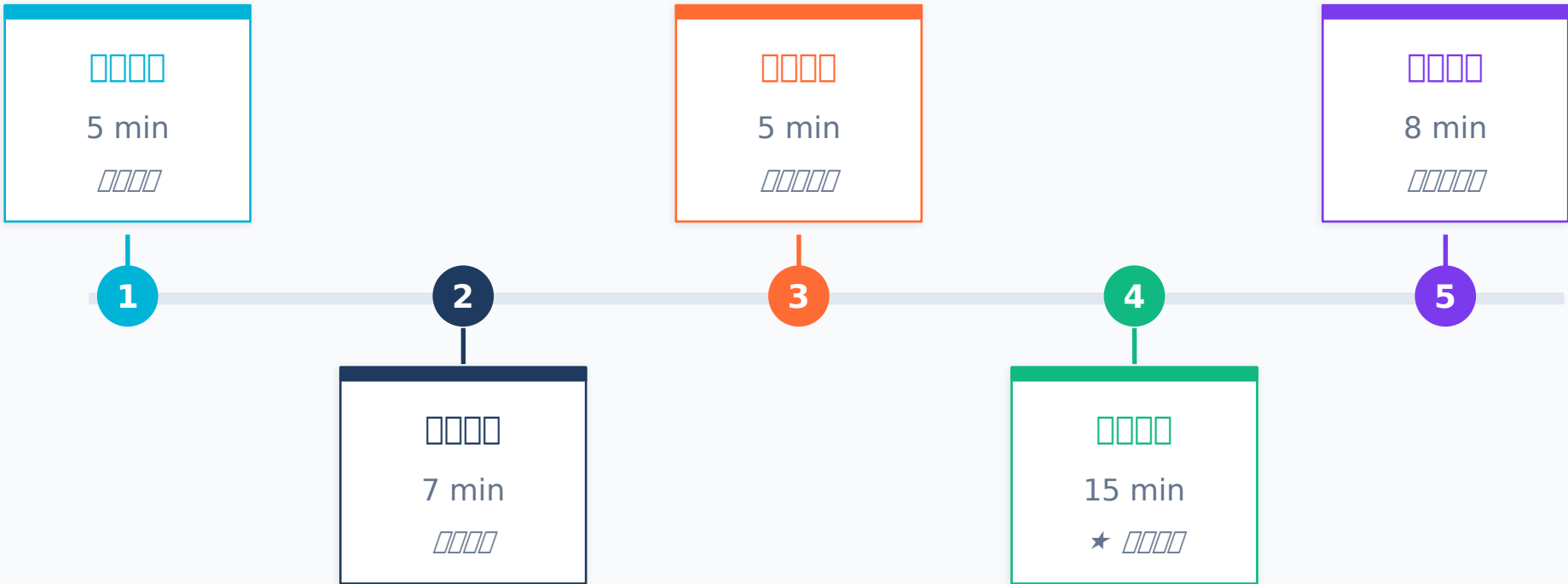
工程 · 設計

- 理解工程設計的過程
- 理解工程設計的原則

資訊科技

- 理解 If-Then 邏輯
- 理解數據的處理

□□□□□□ — 40 □□□□□



□□□□□□ • Teaching Materials

□□□ × 20

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- □□□ > 25°C □□□
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Design Highlights

Unplugged

[illegible]

□ □ □ □ Roleplay

[illegible]

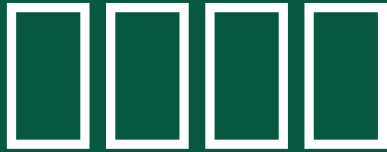
□ □□□ Gamification

33 + 27

□ □□□ Data-Led

The diagram illustrates a transformation process. It starts with a horizontal row of 16 small squares, representing a 16-bit vector. An arrow points from this row to a second row of 4 squares, representing a 4-bit vector. Another arrow points from the 4-bit vector to a third row of 8 squares, representing an 8-bit vector.

PART 04



Classroom Activities · 35% · ~6 min

04

2 □□□□□□□□ / □□□□□□□□□□□□□□□□

4

11

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02 ■ 00000000

7 min

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0000 Dashboard 000000 / 00

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000000

00000000 20-35°C

000000000000

0000000000 90%

000000000000

3. 决策树

5 min

1. 决策树



Root Node

Root Node

温度 > 25°C



Branches

Branches

是 / 否



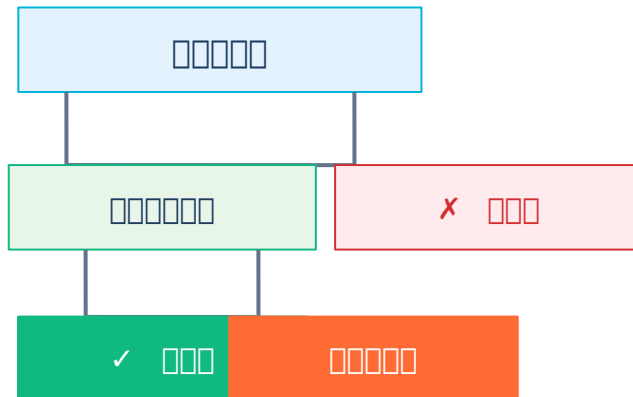
Leaf Node

Leaf Node

是 / 否

决策树

决策树



决策树 If-Then 决策树

第 4 回 環境問題と経済

15 min ★

① 環境

環境 2 環境問題
20 環境問題

② 環境

環境 / 環境問題
= 環境問題↑

③ 環境

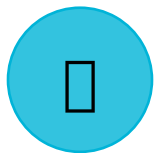
環境問題
環境問題

④ 環境

環境問題 → 環境
環境

環境 5 環境問題

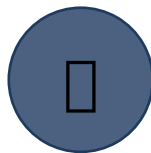
環境



GROUP **A**

DATA & MODEL

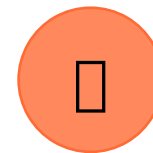
- UCI DATASETS
- Dashboard BUILD
- EDA BUILD



GROUP **B**

MODEL BUILD

- DATASET PREP
- XGBoost BUILD
- EVALUATION



GROUP **C**

DATA & MODEL

- STEM DATASETS
- DATASET PREP
- EVALUATION



Thank you for listening

17K+ 00000000

0 000 · 5 0 ML 0000

0 00 STEM 00000000

0 5 00000000

Q & A · 0000